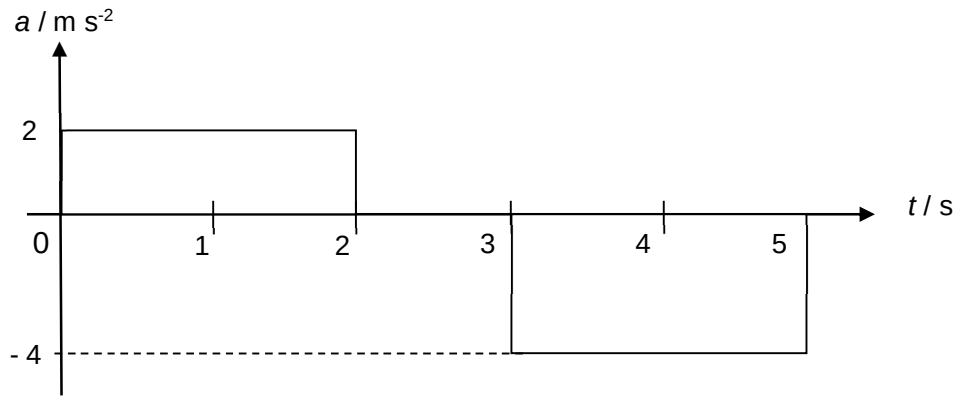


- 4 The graph below represents the variation with time t of the acceleration a of a car starting from rest.

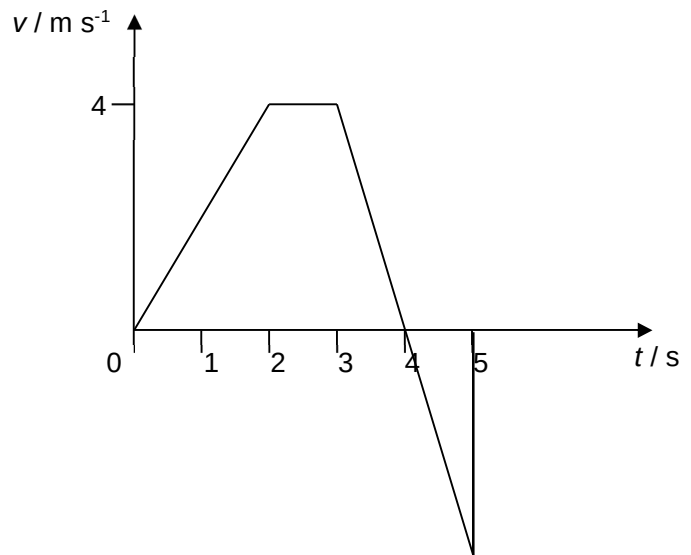


What is the total displacement of the car from the starting point at the end of 5 s?

- A** -4 m **B** 4 m **C** 8 m **D** 12 m

Ans: C

change in displacement
 = Area under the v-t graph
 = $\frac{1}{2} (1+4)(4) - \frac{1}{2} (1)(4)$
 = 8 m



6

5 A rigid cross shaped structure having four arms DO , CO , BO and AO each 1.00 m long is